

ScamSphere AI

AI-Powered Fraud Campaign Intelligence Platform

1. Abstract

ScamSphere AI is an AI-powered fraud campaign intelligence platform designed to detect, analyze, and correlate scam complaints. Traditional scam detection systems focus on identifying whether an individual complaint is fraudulent. However, they often fail to identify relationships between multiple complaints that belong to larger coordinated scam campaigns.

ScamSphere AI addresses this challenge by combining Optical Character Recognition (OCR), Large Language Models (LLMs), campaign correlation techniques, network visualization, and an AI-powered cybercrime copilot into a unified platform.

The system enables users to upload scam screenshots or provide complaint text, automatically extracts relevant intelligence indicators, generates threat assessments, identifies links to historical complaints, visualizes fraud networks, and provides actionable guidance.

2. Problem Statement

Cybercrime incidents such as digital arrest scams, phishing attacks, fake KYC updates, and financial frauds have increased significantly.

Current challenges include:

- Scam reports are analyzed individually.
- Fraud campaigns remain hidden across multiple complaints.
- Citizens struggle to understand scam severity.
- Investigators lack simple tools for identifying relationships between complaints.

As a result, organized scam operations can continue for extended periods before patterns become visible.

3. Proposed Solution

ScamSphere AI transforms isolated scam complaints into actionable cyber intelligence.

The platform performs:

1. Screenshot Analysis using OCR
2. Entity Extraction using AI

3. Threat Assessment
4. Campaign Correlation
5. Fraud Network Visualization
6. AI-Powered Cybercrime Assistance

This enables both citizens and investigators to identify emerging fraud campaigns more effectively.

4. System Architecture

Workflow:

User Input ↓ OCR Engine ↓ Entity Extraction Agent ↓ Risk Analysis Agent ↓ Campaign Correlation Engine
↓ Fraud Network Generator ↓ AI Copilot

Input Sources:

- WhatsApp Screenshots
- SMS Screenshots
- Email Screenshots
- Manual Complaint Text

Output:

- Threat Score
 - Severity Assessment
 - Scam Classification
 - Campaign Links
 - Fraud Network Graph
 - Citizen Guidance
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5. Technology Stack

Frontend:

- Streamlit

Artificial Intelligence:

- Groq API
- Llama 3.3 70B

OCR:

- EasyOCR

Graph Analysis:

- NetworkX

Visualization:

- Plotly

Programming Language:

- Python
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6. Core Modules

6.1 OCR Module

Purpose: Extract textual information from uploaded screenshots.

Technology: EasyOCR

Input: Screenshot Image

Output: Machine-readable text

Example:

Input: WhatsApp screenshot

Output: "CBI has issued an arrest warrant. Transfer Rs 50000 immediately."

6.2 Entity Extraction Module

Purpose: Identify critical indicators from complaints.

Extracted Fields:

- Phone Numbers
- UPI IDs
- Authorities Mentioned
- Monetary Amounts
- Scam Type
- Summary

Technology: Groq Llama 3.3

Benefits: Transforms unstructured complaints into structured intelligence.

6.3 Threat Scoring Module

Purpose: Calculate the severity of complaints.

Outputs:

- Threat Score
- Severity Level
- Risk Factors

Example:

Threat Score: 95 Severity: Critical

Reason: Authority impersonation and urgent payment demand.

6.4 Campaign Correlation Engine

Purpose: Identify links between complaints.

Method:

Compare:

- Phone Numbers
- UPI IDs
- URLs
- Extracted Indicators

If multiple complaints share indicators, they are linked together.

Example:

Complaint A: UPI: fraud@ybl

Complaint B: UPI: fraud@ybl

Result: Campaign Link Identified

This module represents the key innovation of ScamSphere AI.

6.5 Fraud Network Visualization

Purpose: Provide investigators with a graphical representation of relationships.

Visualized Entities:

- Complaints
- Phone Numbers
- UPI IDs

Benefits:

- Identify shared indicators
- Discover scam clusters
- Improve investigation efficiency

Technology: NetworkX + Plotly

6.6 AI Cybercrime Copilot

Purpose: Provide actionable guidance to users.

Example Questions:

- Is this a scam?
- Should I transfer the money?
- What should I do next?
- How can I report this?

Technology: Groq Llama 3.3

Benefits: Makes cybercrime intelligence understandable to non-technical users.

7. Implementation Details

The prototype currently uses a complaint repository containing historical complaint records.

These records simulate real-world complaint databases and enable campaign correlation.

In a production deployment, this repository can be replaced with:

- National Cyber Crime Portal databases

- Banking fraud databases
 - Telecom complaint repositories
 - Law enforcement intelligence systems
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8. Results

The platform successfully demonstrates:

- ✓ Screenshot-based scam analysis
- ✓ Threat scoring
- ✓ Scam classification
- ✓ Complaint correlation
- ✓ Fraud network visualization
- ✓ AI-powered guidance

The prototype successfully identifies connections between complaints that share common fraud indicators.

9. Future Scope

Potential enhancements include:

- Real-time complaint ingestion
 - Voice call transcription support
 - Multi-language analysis
 - Integration with cybercrime portals
 - Automated alert generation
 - Advanced fraud campaign prediction
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10. Conclusion

ScamSphere AI extends traditional scam detection by introducing fraud campaign intelligence.

Instead of treating complaints as isolated events, the platform correlates reports, uncovers relationships, visualizes fraud networks, and provides actionable recommendations.

The solution demonstrates how artificial intelligence can transform raw complaint data into meaningful cybercrime intelligence for citizens and investigators alike.